

Backpropagation and Biologically Plausible Gradient Descent Algorithms

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Abstract. Recent advances in artificial intelligence and deep learning have been primarily driven by the backpropagation algorithm. The biological plausibility remains questionable to its reliance on implausible mechanisms such as global error signals, weight-symmetry, and phase-dependent updates. This review examines recent developments in narrowing the gap between artificial and natural learning paradigms that are considered as plausible substitutes for backpropagation. We assess a variety of algorithms particularly Hebbian learning, spike-timing dependent plasticity, target propagation, predictive coding, dendritic computations and localized learning, feedback alignment variants, energy-based learning, and the forward-forward algorithm. Our comparative analysis of these methods shows that although substantial progress has been made in developing bio-inspired algorithms, challenges remain in matching the computational efficiency of traditional backpropagation while maintaining said biological plausibility. Hence, our in-depth review also highlights future research avenues toward developing well-informed brain-inspired learning methods.

Keywords: Backpropagation, Bio-inspired, Deep learning, Hebbian Learning

1 Introduction

Backpropagation [1] has played a crucial role in the rapid advancement of deep learning predominantly used under the setting of supervised learning environments. The algorithm utilizes versions of gradient descent to iteratively modify the weights of a deep network and regress the network's output to a provided label, achieving almost perfect performance on complex cognitive tasks such as image and language understanding. Despite its clear success and widespread use in various fields, the biological validity of backpropagation continues to be a controversial topic among neuroscientists and computer scientists alike.

The main objections to the biological relevance of backpropagation arise from three needs of the algorithm: the need of a two-phase training process that includes separate forward and backward steps, the need for symmetrical weight adjustments during both phases of training, and the need to calculate global errors that must be transmitted through all layers. These characteristics of backpropagation have no clear counterparts

in neurobiological systems which function within the boundaries of localized information processing.

Acknowledging these shortcomings, researchers have made notable progress in developing alternative training methods that aim to better reflect biological mechanisms, such as Hebbian learning [2,3,4], Spike-Timing Dependent Plasticity (STDP) [5], target propagation [6], predictive coding [7,8], dendritic computations and localized learning [11,12], feedback alignment [13, 14, 15,16], energy-based learning [20,21], and the forward-forward algorithm [22]. Each of these methods provides a distinct viewpoint on how modifications in synaptic networks could occur in a way that aligns with the biology of the brain.

Our contributions can be summarized as:

- We conduct a critical analysis of backpropagation and discuss its biological implausibility.
- We review the current landscape of biologically plausible algorithms by explaining each of the aforementioned methods.
- We provide a comparative analysis to elucidate computational efficiency of these methods.
- We conclude by acknowledging the drawbacks of these methods and emphasizing potential future research directions aimed at developing informed brain-inspired learning methods.

2 A Critical Analysis of Backpropagation

Backpropagation, as established by Rumelhart, Hinton, and Williams [1], was the foundation of deep learning because it enabled the optimization of the weights efficiently using gradient descent. Backpropagation, in the traditional formulation, estimates the error at the output layer and then propagates the errors backward through the network using the chain rule. More precisely, for a network with layers indexed by l , the error signal at each layer is computed as

$$\delta_l = f'(z_l) \cdot W_{l+1}^T \delta_{l+1}, \quad (1)$$

Where z_l denotes the pre-activation at layer l , $f'(\cdot)$ is the derivative of the activation function, and W_{l+1}^T is the weight matrix connecting layers l and $l+1$. Although this algorithm has proven extremely effective for training artificial neural networks, it relies on mechanisms not readily compatible with biological learning. First, the algorithm requires a two-stage process—a forward pass stage and a backward pass stage—implying the need for an extra pathway for error signals. Second, it requires symmetric weight matrices (i.e., identical weights for forward and backward passes), a requirement dubbed the “weight transport problem” with no obvious analogue in the brain [16]. Third, the global error signal propagation through many layers is incompatible with the locally dominated nature of synaptic plasticity observed in biological networks, where updates are dependent on local signals such as spike-timing [5]. Thus, while backprop-

agation is computationally efficient for artificial networks, its lack of biological plausibility has spurred the development of alternative learning algorithms that adhere more closely to neurobiological constraints.

2.1 Biological Implausibility of Backpropagation

Backpropagation, as introduced by Rumelhart, Hinton, and Williams [1], has led the pack in modern deep learning due to its remarkable success in multilayer network learning. However, the algorithm as computationally efficient as it is remains largely considered biologically implausible on a number of basic reasons. One of these is the weights transport problem: backpropagation requires the exact transposition of the forward weight matrices during the backward pass, an implication of symmetry of connectivity of neurons not present in biological neural networks [4]. Synaptic relationships in biology are primarily one-way and do not possess the exact reciprocal structure backpropagation assumes.

Another critical issue is the two-phase learning procedure dependence of the algorithm. Traditional backpropagation separates computation into distinct forward and backward passes, with the forward pass computing the activations and the backward pass backpropagating global error signals to update the weights. In contrast, the biological neurons compute in a continuous, highly parallel manner, and local synaptic plasticity mechanisms are responsible for learning. Experimental evidence suggests that synaptic plasticity in the brain is triggered by local signals, such as those dictated by spike-timing-dependent plasticity (STDP), where the precise timing of pre- and post-synaptic spikes dictates synaptic strength changes [8]. This local mechanism is in stark contrast to the requirement of global error backpropagation across multiple layers in artificial networks.

Also, backpropagation involves computation and backpropagation of global error signals informing each neuron its contribution to the overall loss. In the biological example, it is not clear whether such exact error signals are backpropagated through the complex and highly distributed brain networks. Instead, the brain implements more local types of error correction, as predictive coding theories propose, where neurons are constantly updating their activity based on locally computed prediction errors without such clear backpropagation of a global error [7,8].

These discrepancies highlight the fundamental incompatibility between the operations of backpropagation and those of biological learning systems. Research like that of Lillicrap et al. [16] has brought these problems to rigorous critical attention, highlighting that although backpropagation is a powerful artificial system algorithm, its fundamental assumptions are incompatible with the fact of neurobiology. Therefore, there is considerable interest in the design of different learning algorithms—like Equilibrium Propagation and Feedback Alignment—closer to the constraints and operating principles of biological networks.

3 Biologically Plausible Learning Algorithms

3.1 Hebbian Learning

Hebbian learning is a key principle in understanding how neurons in the brain adapt and learn from their experiences. The concept, put forth by Donald Hebb, posits that the connection between two neurons strengthens when they are activated together repeatedly, often encapsulated in the phrase, “neurons that fire together, wire together.” The idea is deemed biologically plausible since it relies solely on the inputs received by the system to create patterns of activity, without the need for explicit teaching signals.

In the context of artificial neural networks, Hebbian learning is realized by adjusting the connection weights between two interacting units based on the activity levels of the connected units, or neurons. The fundamental Hebbian learning rule is mathematically expressed as:

$$\Delta w = \eta y(x, w)x \quad (2)$$

In this equation, Δw signifies the change in weight, a non-linear function $y(x, w)$ is the post-synaptic activation of the neuron η , which determines the speed at which the weights change in response to the activation they receive. The activation level of a neuron typically lies between the range of 0 to 1. The drawback of this rule is that it only allows unbounded accumulation of the weights. To prevent this, a weight decay term proportional to the value of the weight is introduced to normalize the gradient, given by:

$$\Delta w = \eta y(x, w)(x - w) \quad (3)$$

This adjustment ensures that the weight vector is updated with the movement closer to the input after each iteration, with the step size being proportional to the similarity between the input and weight vector. Over time, if a specific input or a group of similar inputs is presented repeatedly, the weight vector converges toward it. This process allows the neuron to function as a matching filter and “memorize” the input in its synaptic weights.

The versatility of Hebbian learning makes it applicable across various neural network architectures and learning tasks. HebbNet [25] is a neural network that enhances the algorithm by incorporating updated activation thresholds and gradient sparsity. Similarly, SoftHebb [26] is another algorithm that merges deep learning elements with Hebbian plasticity, providing an unsupervised Bayesian model that can generate synthetic objects as interpolations of real object classes.

3.2 Synaptic Time-Dependent Plasticity

In biological neural networks, the pre-synaptic neuron refers to the neuron that sends the signal, while the post-synaptic neuron is the recipient of the signal. The “spike” is

the rapid change in voltage across the neuron's membrane known as action potential. The synaptic weight indicates the strength or influence of connection between neurons. Spike-Timing Dependent Plasticity or STDP is a learning rule within Hebbian learning models that adjusts synaptic weights based on the timing of spikes. If a post-synaptic spike closely follows a pre-synaptic spike, it suggests a strong causal relationship between the two neurons' activities. In such cases, the synaptic weight is increased to strengthen the connection. The shorter the time interval between the spikes, the stronger the inferred relationship, leading to a greater increase in weight. This is called long-term potentiation (LTP). If the post-synaptic spike occurs before the pre-synaptic spike, the synaptic weight is reduced, leading to long-term depression (LTD). The rule is expressed mathematically as follows:

$$\Delta w = \alpha + * \exp(\Delta t / \tau+) \text{ if } \Delta t > 0 \quad (4)$$

$$\Delta w = \alpha - * \exp(\Delta t / \tau-) \text{ if } \Delta t < 0, \quad (5)$$

Here, Δt represents the time difference between the post-synaptic spike (t_{post}) and the pre-synaptic spike (t_{pre}). The parameters $\alpha+$ and $\alpha-$ determine the maximum synaptic modification, while $\tau+$ and $\tau-$ are time constants that control the rate of decay for LTP and LTD, respectively.

The biological mechanisms that STDP is based on are closely tied to calcium Ca^{2+} dynamics in the post-synaptic neuron [27,28]. These calcium levels activate distinct molecular pathways: high Ca^{2+} levels activate CaMKII for LTP, while moderate levels activate PP1 and calcineurin for LTD, while moderate levels activate PP1 and calcineurin for LTD. The amount of Ca^{2+} influx depends on the timing between the excitatory postsynaptic potential (EPSP), caused by the pre-synaptic spike, and the backpropagating action potential (BAP), triggered by the post-synaptic spike. Positive intervals (BAP following EPSP) result in a substantial Ca^{2+} influx, while negative intervals (BAP preceding EPSP) lead to a moderate influx.

3.3 Target Propagation

Target Propagation (TP) is a class of biologically inspired learning algorithms that have been put forward to overcome the key shortcomings of backpropagation regarding biological plausibility. TP, as presented by Bengio [6], circumvents the computation and explicit error gradient propagation and propagates target activations through the network layers. Targets are the values that each layer ought to take in order to minimize the overall output error. Each layer then modifies its weights according to local loss functions that push its actual activation towards this target, thereby performing a local approximation to gradient descent.

The main innovation is that it eliminates the necessity for transposition of forward weights (resolving the weight transport problem) and does not backpropagate global error signals. Rather, locally calculated activation targets are utilized to guide learning, which is more biologically plausible in layered cortical networks.

In a standard feedforward network, let:

- x^l be the activation of the l -th layer,
- W^l the forward weights connecting layer $l-1$ to l .
- $f^l(\cdot)$ the nonlinear transformation at layer l ,
- And $\hat{f}^l(\cdot)$ the approximate inverse function of f^l (learned by training a decoder or inverse network).

The forward pass computes activations layer by layer:

$$x^l = f^l(W^l x^{l-1}). \quad (6)$$

Unlike backpropagation, which computes gradients for each layer's weights using the chain rule, target propagation calculates a target activation for each hidden layer, which approximates the activation that would reduce the output loss. This is computed using:

$$\tilde{x}^{l-1} = \hat{f}^l(\tilde{x}^l), \quad (7)$$

where $\tilde{x}^l$ is the target for layer l , computed recursively starting from the output layer.

At the output layer L , we define:

$$\tilde{x}^L = x^L - \eta \frac{\partial L}{\partial x^L}, \quad (8)$$

where L is the loss function (e.g., mean squared error), and η is the learning rate.

Each hidden layer's weights W^l are then updated by minimizing a local loss function:

$$\epsilon^l = ||x^l - \tilde{x}^l||^2. \quad (9)$$

This means that each layer learns to produce activations that match their respective targets, computed using the approximate inverse mappings.

Target propagation enhances biological plausibility as it eliminates symmetric weights; the inverse functions are learned separately, bypassing the weight transport issue. Learning is local, and every layer minimizes its loss depending on the difference between its activation and a target, without any global error signals. The approach provides hierarchical credit assignment via recursive target computation, which is analogous to top-down signaling in the brain. Its dependence on alternating forward and

backward phases renders it not fully concurrent, somewhat decreasing its biological realism, although this is ameliorated in variants such as Difference Target Propagation.

3.4 Feedback Alignment

Feedback Alignment (FA) is a learning algorithm designed to overcome one of the key challenges of traditional backpropagation in artificial neural networks. In conventional backpropagation, error signals are computed at the output and then sent backwards through the network using the exact transposition of the forward weight matrices. This method requires the network to maintain perfectly symmetric connections between layers, a condition known as "tied" weights. This symmetric connectivity, however, is not observed in biological neural networks, which display heterogeneous and largely unidirectional synaptic connections.

FA offers a biologically plausible alternative to traditional backpropagation by addressing its most problematic requirement: the need for symmetric, or "tied," weights between the forward and backward passes.

In FA, the algorithm replaces the transposed weight matrix with fixed, random feedback matrices. Consider a feedforward neural network with L layers. During the forward pass, activations are computed in the usual manner:

$$a_i = \sigma(z_i) \text{ with } z_i = W_i a_{i-1}, \quad (10)$$

where $\sigma(\cdot)$ is the activation function and W_i is the weight matrix connecting layer $i-1$ to layer i .

In traditional backpropagation, the error signal at layer i is given by:

$$\delta_i = \sigma'(z_i)(W_{i+1}^T \delta_{i+1}), \quad (11)$$

with the output error defined as:

$$\delta_L = \sigma'(z_L)(y - a_L), \quad (12)$$

Feedback Alignment modifies this computation by substituting the transposed weight matrix $W_{i+1}^T \delta_{i+1}$ with a fixed random matrix B_{i+1} :

$$\delta_i = \sigma'(z_i)(B_{i+1} \delta_{i+1}). \quad (13)$$

The weight update rule remains similar to standard backpropagation:

$$\Delta W_i = -\eta \delta_i a_{i-1}^T, \quad (14)$$

Where η is the learning rate.

This modification eliminates the requirement for exact reciprocal connections between layers, aligning more closely with the observed biological constraints where precise backward connectivity is unlikely. Instead of propagating a calculated global error signal, each layer receives an approximate error through fixed, random projections. Despite this approximation, experimental studies have demonstrated that networks trained with FA can still converge and perform competitively [14].

From a biological perspective, FA is piques our interest because it mirrors the decentralized nature of synaptic connectivity. The brain does not appear to employ exact mirror-image connections for error transmission; rather, local synaptic modifications are driven by noisy and approximate signals. Feedback Alignment thus supports the idea that the brain might use local, approximate error signals delivered via random or heterogeneous pathways to guide learning and adaptation. This property makes FA computationally attractive and also provides a plausible mechanism by which biological systems could implement learning without the strict constraints imposed by traditional backpropagation.

3.5 Predictive Coding

Predictive coding is a theoretical framework that postulates the brain as a hierarchical prediction machine. According to this view, each level in the cortical hierarchy continually makes predictions about the activity in the level below and computes a prediction error—the difference between the actual input and the prediction. These local errors are then used to update both the representations (activations) and synaptic weights, effectively minimizing the overall prediction error in a manner analogous to gradient descent. This framework is considered a biologically plausible way to implement error-driven learning because it relies solely on local computations and does not require global error signals or symmetric weight matrices.

Two seminal works that gave rise to this framework are Rao and Ballard [8] and Whittington and Bogacz [4]. Rao and Ballard's paper laid the groundwork by demonstrating how predictive coding could account for extra-classical receptive-field effects observed in the visual cortex. Whittington and Bogacz further developed the idea, showing how predictive coding networks can approximate the error backpropagation algorithm using local Hebbian plasticity rules. In a typical predictive coding network, there is a hierarchy of layers' $l = 0, 1, \dots, L$:

- **Value Neurons (x^l):** These represent the estimates or predictions at layer l .
- **Error Neurons (ϵ^l):** These compute the local prediction error at layer l as the difference between the actual input received from the lower layer and the top-down prediction from the higher layer.

For example, assume that the network is organized such that the activity x^l of the value neurons in a given layer is used to predict the activity x^{l-1} in the layer below through a nonlinear function f^l and weight matrix W^l . The top-down prediction for layer $l-1$ is given by:

$$\mu^{l-1} = f^l(W^l x^l). \quad (15)$$

The local prediction error at layer $l-1$ is then defined as:

$$\epsilon^{l-1} = x^{l-1} - \mu^{l-1}. \quad (16)$$

This error is used locally to update the activities and the weights. The error signals are propagated upward while the predictions are sent downward.

Predictive coding is done through local computation in which every neuron updates its state solely based on the difference between the top-down predictions and feedforward sensory input without requiring any global error signal. It prevents symmetric weights between feedforward and feedback pathways like backpropagation and thereby prevents the biologically implausible weight transport issue. Furthermore, both inference and learning are performed in predictive coding networks in parallel with continuous dynamics, which is equal to the real-time and parallelism of neural computation in biology. Finally, the model possesses a probabilistic interpretation within the free energy principle, where the minimization of prediction errors corresponds to approximate Bayesian inference under uncertainty.

3.6 Dendritic Computations for Local Learning

Backpropagation's biological implausibility motivated numerous brain-inspired learning rules. Cortical pyramidal neurons with compartmentalized dendrites, for instance, inspire a local-error learning framework. Pyramidal cells take feedforward (sensory) inputs on basal dendrites and modulatory/top-down inputs on apical dendrites [29, 30]. Compartmentalization enables each neuron to calculate a local error by comparing its bottom-up input with a top-down "prediction" or target at the soma. For instance, Fischer et al. describe that bottom-up sensory inputs synapse onto proximal (basal) compartments and long-range feedback (contextual or reward-prediction signals) onto apical tuft dendrites [29, 30]. Local error for a 3-compartment neuron like this can be written as

$$e_i = z_i - \hat{z}_i, \quad (17)$$

where z_i is the actual sensory drive (basal input) and $\hat{z}_i$ is the "expected" value from feedback (the apical input) [29]. Thus each neuron uses only its own feedforward activation and feedback signal to measure an error, in contrast to global error backpropagation. Stacking layers of such units yields a local layerwise error

$$E_l = \frac{1}{2} \sum (z_i^l - \hat{z}_i^l)^2, \quad (18)$$

and the total network loss $E_l = \frac{1}{2} \sum (z_i^l - \hat{z}_i^l)^2$, [1]. Crucially, DLL abandons weight symmetry by using separate feedback weights Θ^l (trained independently of the forward weights W^l), meeting the “no weight transport” criterion [29]. Likewise, feedforward and feedback signals can coexist without strict phases, since basal and apical inputs are spatially distinct [29].

Prior architectures have employed similar concepts. Guerguiev et al. (2017) showed that the ability to have distinct basal vs. apical compartmentalization within neurons allows for deep credit assignment: feedforward stimuli hit one compartment and feedback another to allow for multilayer learning without symmetrically coupled weights [29, 31]. Adding interneuron-mediated apical “prediction errors” to initiate synaptic plasticity, Sacramento et al. (2018) showed analytically that dendritic errors locally approximate backpropagation [11]. More general classes of dendritic models with local adaptivity (e.g. Hebbian or burst-dependent STDP) can more broadly perform credit assignment [33, 11]. The studies show that dendrites offer a two-layered organization in one neuron that naturally allows errors from the soma to be backpropagated to the synapses [32, 11]. For instance, Urbanczik et al. (2016) derived a plasticity rule whereby apical feedback controls error-correcting updates to basal synapses [32, 11].

DLL Algorithm Formulation

Network and Neuron Model: DLL models each hidden unit as a three-compartment pyramidal neuron (basal dendrite, apical dendrite, soma) [29]. During training, feedforward activations a^l propagate through weights W^l from layer $l-1$ to l (with any activation function f), while feedback signals $\hat{z}^l$ propagate downward through separate weights Θ^l from layer $l+1$ to l . The local expected value at layer l (the apical signal) is iteratively computed by differentiating the loss of layer $l+1$ through the feedback weights. In practice, DLL defines a “backpropagated activity” $\hat{z}^l$ at each layer that satisfies

$$\hat{z}^l = f'(a^l) \odot (\hat{z}^{l+1} \Theta^{l+1}), \quad (19)$$

Where $\odot$ is elementwise product [29]. At the top (output) layer, $\hat{z}^L$ is simply the mismatch $(z^L - y)$ times the derivative $f'(a^L)$ (for MSE loss), analogous to backpropagated gradient. In the first epoch (or initialization), layers below the top simply set $\hat{z}^l$ equal to the feedforward activation (so that the network initially “predicts” its own output) [29]. Over training, $\hat{z}^l$ continually reflects the target-directed error at each layer via the apical (feedback) input.

Local Error and Loss: By construction, each neuron at layer l compares its basal drive z^l to the apical prediction $\hat{z}^l$. The local loss for layer l is therefore

$$E_l = \frac{1}{2} \sum (z_i^l - \hat{z}_i^l)^2, \quad (20)$$

summed over units i in layer l [29]. The global objective is $L = \sum_l E_l$ [1]. (Note that under this scheme, minimizing $\sum_l E_l$ aligns with traditional MSE on the output, but through layerwise terms.) This local error formulation avoids any global error propagation beyond the dendritic feedback that carries $\hat{z}^l$.

Weight Updates: DLL performs (stochastic) gradient descent on the local loss at each layer using only local variables. In particular, each forward weight W_{ij}^l (from neuron j in layer $l-1$ to i in layer l) is updated by

$$\Delta W_{ij}^l = -\eta_w \frac{\partial E_l}{\partial W_{ij}^l} = -\eta_w (z_i^l - \hat{z}_i^l) f'(a_i^l) a_j^{l-1}. \quad (21)$$

where a_j^{l-1} is the pre-synaptic activation and $(z_i^l - \hat{z}_i^l) f'(a_i^l) = \xi_i^l$ is the post-synaptic local error term [29]. Similarly, each feedback weight Θ_{ij}^l (from neuron i in layer l to j in layer $l-1$) is updated by its own local error:

$$\Delta \Theta_{ij}^l = -\eta_\theta \frac{\partial E_{l-1}}{\partial \Theta_{ij}^l} = -\eta_\theta (z_j^{l-1} - \hat{z}_j^{l-1}) f'(a_j^{l-1}) a_i^l. \quad (22)$$

In practice, the DLL paper expresses these updates compactly and applies them simultaneously for all layers [29]. Crucially, each update depends only on the activity of the connected pre- and post-synaptic neurons and their difference (the local error), satisfying strict locality. There is no need to wait for a global backward phase: feedforward and feedback signals coexist, and W^l and Θ^l are adjusted concurrently at each training step [29].

Approximation of Backpropagation

Despite its locality, DLL approximates gradient descent on the overall loss. Intuitively, when apical inputs $\hat{z}^l$ correctly convey the downward gradient signal, the sum of all layerwise updates is equivalent to full backprop. Lv et al. demonstrate that when training converges, the fixed points of DLL satisfy $\hat{z}^l = z^l$ (no local error), paralleling the condition that the network output equals the target. Analytically, by substituting $\hat{z}^l$ into the updates, one finds that DLL's weight changes follow the same form as $-\frac{\partial L}{\partial W^l}$ using separate feedback weights [29]. Empirically, DLL achieves performance on par with backprop-based training across MLPs, CNNs, and RNNs [29, 11]. For example, Lv et al. report that DLL attains state-of-the-art accuracy among biologically plausible algorithms on benchmarks (MNIST, CIFAR-10, etc.) while meeting all three plausibility criteria [11, 29].

Dendritic Localized Learning leverages biologically inspired mechanisms—compartmentalized neurons and local Hebbian-like updates—to approximate gradient descent

without non-local signals. Each neuron computes a local error from the difference between basal (sensory) drive and apical (predictive) feedback [29]. The resulting learning rule uses separate feedback weights and concurrent updates, thus satisfying biological constraints (asymmetry, locality, no temporal phasing) while effectively training deep networks [29, 11]. This approach illustrates how dendritic processing could endow neural circuits with a practical credit-assignment mechanism analogous to backpropagation [32, 11].

3.7 Energy-Based Learning: Equilibrium Propagation

One such strongest framework of biologically plausible gradient descent is Equilibrium Propagation (EP), proposed by Scellier and Bengio [34]. EP is a technique of training energy-based models (EBMs) with more biologically plausible neural computation dynamics compared to the traditional backpropagation. Contrasting the traditional neural networks that compute gradients directly via backpropagation, EP utilizes local neural dynamics to estimate the gradient descent via energy minimization in two steps.

Theoretical Framework

EP is built upon energy-based models, where the system's state is defined by a set of variables $E_l = \frac{1}{2} \sum (z_i^l - \hat{z}_i^l)^2$, with h being the hidden units and y the output units. The system's behavior is governed by an energy function:

$$E_l = \frac{1}{2} \sum (z_i^l - \hat{z}_i^l)^2, \quad (23)$$

where θ are the network parameters and x is the input. The system evolves to minimize this energy via gradient descent dynamics:

$$\frac{ds}{dt} = -\frac{\partial E}{\partial s}, \quad (24)$$

which represents a continuous relaxation process akin to the leaky integrator models of biological neurons. At equilibrium, the state s minimizes the energy given fixed parameters and input.

To perform learning, EP introduces a perturbed energy function:

$$F(\theta, \beta, s) = E(\theta, x, s) + \beta C(s, y), \quad (25)$$

where $C(s, y)$ is a cost function measuring the error between output and target (such as mean squared error), and β is a small nudging parameter that modulates the influence of the cost function. When $\beta = 0$, the system is said to be in the free phase, and when $\beta > 0$, the system is in the nudged phase [34].

Learning Dynamics

EP operates in two stages. First, in the free phase ($\beta = 0$), the system reaches a stable state s^0 by minimizing the energy function without the influence of the target. Second, in the nudged phase ($\beta > 0$), the system state s^β evolves under a small influence of the cost function, which gently pushes the output towards the target.

The gradient of the loss with respect to parameters θ is approximated by:

$$E_1 = \frac{1}{2} \sum (z_i^1 - \hat{z}_i^1)^2, \eta/\beta \left(\frac{\partial E}{\partial \theta} (\theta, x, s^\beta) - \frac{\partial E}{\partial \theta} (\theta, x, s^0) \right), \quad (26)$$

where η is the learning rate. This learning rule can be derived as an approximation of stochastic gradient descent, with accuracy improving as $\beta \rightarrow 0$. This update uses only locally available signals and does not require backpropagation of gradients through the network, making it far more plausible from a biological perspective [34].

Mathematically, it has been shown that:

$$\Delta\theta = -\eta E_1 = -\frac{1}{2} \sum (z_i^1 - \hat{z}_i^1)^2, \quad (27)$$

which confirms that equilibrium propagation approximates true gradient descent up to an error of order β [34].

Biological Plausibility

EP satisfies several significant biological requirements. Learning is completely local and relies only upon pre- and postsynaptic states for every synapse. The system evolves continuously, similar to biological time development, and doesn't involve alternation between computation's forward and reverse processes. Employing bidirectional and symmetric weights for layered networks is reminiscent of the predominance of two-way inter-connections among cortical microcircuits [34, 35].

Its dynamics are associated with biophysical processes such as conductance-based leaky integrate-and-fire models [36], and its learning rule corresponds to spike-timing-dependent plasticity (STDP), indicating its relevance to neurobiology. Because of the similarity between EP and temporally observed learning and processes at the network level in biology, EP is one of the best candidates for learning that is biologically realistic.

Applications and Extensions

EP is generalized further in several directions. Holomorphic Equilibrium Propagation removes independent forward and backward phases by complex-valued dynamics and periodic modulation such that precise gradient computation is achievable by using finite-size oscillations [35]. EP also achieves successful implementation in hardware models such as memristor-based recurrent neural networks [37], which demonstrate its compatibility for future neuromorphic systems.

EP has also been generalized to quantum systems, where quantum Hamiltonians are used as expectation values for the energy function and its usage generalized to quantum machine learning [38]. EP is also robust to both noisy conditions and hardware fluctuations and its performance is retained even if there are perturbations [39].

Equilibrium Propagation is a principled, biologically realistic, and backpropagation-free alternative to train neural networks. Through energy minimization and soft nudging of the system towards target outputs, EP provides a gradient-descent-compatible framework relying on local computations and biologically realistic dynamics. It is among the most theoretically sound and practically relevant learning frameworks in the current endeavor to unify machine learning and neuroscience.

3.8 Forward-Forward Algorithm

The Forward-Forward (FF) algorithm, introduced by Geoffrey Hinton in 2022 [22], proposes a radically different approach to deep learning that avoids the biologically implausible aspects of traditional backpropagation. The FF approach eliminates the use of the backward pass entirely and uses two distinct passes of forwards - one on positive (real) and one on negative (false or corrupted) data. It attempts to train neural networks by being entirely dependent on local calculations, which better matches biological learning.

Traditional backpropagation requires both non-local error signals and symmetry of weights to attribute credit, and its biological feasibility is thus very difficult. FF is still an attempt to simulate the learning from more neural system-like processes. Rather than using gradients to be computed by propagating error signals backwards, FF is a process only going forwards based on a "goodness" function, local learning signal at synapses [22].

Architecture and Dynamic

In the FF framework, each layer of the neural network receives an input vector and computes a hidden representation. For a layer l , the pre-activation is defined as:

$$z^l = W^l x^l + b^l, \quad (28)$$

where W^l is the weight matrix, b^l is the bias vector, and z^l is the input from the previous layer. The activation is passed through a non-linear function, often ReLU or tanh, to produce the output:

$$h^l = f'(z^l), \quad (29)$$

The goodness of the representation at a given layer is measured using the squared norm of the activations:

$$G^l = \|h^l\|^2, \quad (30)$$

A higher goodness value implies that the layer strongly recognizes or responds to the given input. For training, each sample is labeled as positive (i.e., matching its correct class or real input) or negative (e.g., containing incorrect labels or synthetic/noisy inputs).

Training Procedure

The FF algorithm trains the network by performing two separate forward passes:

- **Positive Pass:** The model processes clean, correctly labeled data and encourages the activations to have high goodness.
- **Negative Pass:** The model processes incorrect or corrupted data and encourages the activations to have low goodness.

The training objective is to increase the goodness for positive samples and decrease it for negative ones. For a single layer, this can be cast as a logistic objective:

$$L = \log(1 + e^{G_{neg} - G_{pos}}), \quad (31)$$

Where G_{pos} and G_{neg} are the total goodness scores for positive and negative examples, respectively [22].

The gradients are computed locally at each layer based on the difference in goodness values. Importantly, the model does not need to propagate gradients through the network or rely on symmetric weight matrices. Each layer updates its weights independently using only its own activations:

$$\Delta W^l \propto \frac{\partial G_{pos}^l}{\partial W^l} - \frac{\partial G_{neg}^l}{\partial W^l}. \quad (32)$$

This local learning rule is biologically plausible, as it resembles Hebbian mechanisms where weight changes depend on correlated pre- and post-synaptic activity.

Biological Plausibility

The FF algorithm fulfills several significant requirements for biological feasibility:

- **No Backward Pass:** The learning relies solely on the forward activations and doesn't involve non-local backward propagating signals.
- **Local Computation:** Each layer only uses its local activity to perform the weight updates.
- **No Weight Symmetry:** There is no necessity for feedback weights to be the same as feedforward weights.

These attributes make FF one of the more realistic biological methods proposed for deep networks to date. Its learning law and architecture also resemble how actual neurons would learn to adjust the strength of synapses from a stream of multiple inputs.

Extensions and Results

Hinton et al. provided evidence that the FF algorithm is just as effective as backpropagation for tasks such as MNIST classification and even for multilayered networks [22]. Follow-up work expanded FF to convolutional networks and to generative models and employed it to demonstrate its versatility and power [40]. The model may be further extended to other regularization techniques, such as contrastive goodness margins, and hybrid models incorporating other methods of unsupervised learning.

The FF algorithm also inspired studies on goodness-based learning across other perceptual modalities, such as vision and hearing, and was further applied to spiking neural networks, once again bridging the artificial-biological divide [40, 41].

Forward-Forward is a dramatic shift from the conventional paradigm of the use of backpropagation. Both its dual-pass architecture and locality of goodness-based learning render FF a learning strategy that is both computationally effective and biologically realistic. It possesses the potential that future brain-inspired intelligent systems may largely depend on processes of forward-only learning given its success across many tasks.

4 Comparative Analysis

We evaluate and compare the biological plausibility of these methods using criteria such as asymmetry of forward and backward weights, local error representation and non-two-stage training.

We observe that Hebbian learning and STDP are the most biologically plausible methods as they avoid backpropagation of errors by relying solely on local information

for reflecting synaptic changes, and do not require symmetric weights. Hebbian learning represents the most basic form of synaptic plasticity, operating on the principle that simultaneous activation of neurons strengthens their connections. While foundational, its simplicity can lead to stability issues and uncontrolled synaptic growth. STDP builds upon Hebbian principles but adds temporal precision, considering the exact timing of pre- and post-synaptic spikes. This temporal dependency makes STDP more sophisticated and biologically accurate than basic Hebbian learning, as it better reflects the actual mechanisms observed in neural systems.

Predictive coding represents a fundamentally different approach, focusing on how the brain generates and updates internal models to predict sensory inputs. Unlike Hebbian learning and STDP, which operate primarily at the synaptic level, predictive coding provides a broader framework for understanding neural processing through hierarchical prediction and error correction. It satisfies only the local error representation criterion for biological plausibility.

Target propagation and feedback alignment both are viable alternatives to the backpropagation algorithm. Target propagation computes local targets for each layer rather than propagating gradients, which helps in avoiding vanishing and exploding gradient issues as well, whereas feedback alignment uses random feedback matrices instead of symmetric weights. Both of these methods satisfy the weight asymmetry and local error representation criterion.

The Forward-Forward algorithm and Energy-Based Learning both meet the three important criteria for alternative learning methods: they avoid the need for symmetric weights, rely on local error signals, and do not require parallelized training phases. The Forward-Forward algorithm simplifies the learning process by merging the forward and backward propagation phases into a single, unified process. This approach aligns more closely with how biological systems are thought to function, making it a more biologically plausible alternative to traditional backpropagation. On the other hand, Energy-Based Learning focuses on minimizing a global energy function while still utilizing local error signals and maintaining asymmetry in the weights.

Dendritic Localized Learning (DLL) [29] is the only algorithm that satisfies all three criteria of biological plausibility. Compared to target propagation, DLL propagates local errors instead of targets, avoiding the issues of class variability that are inherently encountered in TP and its variants, while maintaining asymmetry of forward and backward weights. Similarly, DLL shares the concept of local error computation with predictive coding, however, differs in the case of weight symmetry.

5 Drawbacks and Future Research Directions

The aforementioned methods, while promising, face several significant drawbacks that hinder their progress and widespread adoption. Hebbian learning and STDP, while being the most biologically plausible algorithms, exhibit poor performance on tasks such as image classification as compared to traditional deep learning architectures, particularly on the CIFAR-100 dataset [42]. The Forward-Forward algorithm and Energy-Based Learning, despite their innovations, fall short in addressing certain key aspects

of biological neural systems. Specifically, they do not account for the spiking behavior of biological neurons or the unsigned nature of biological error signals. Target propagation and Predictive coding are algorithms that maintain local error computation, but lack a non-two-stage training approach, affecting their biological plausibility [29].

The field being in its early stages requires several critical research directions to be explored. One major focus is scaling biologically plausible models to handle complex, high-dimensional tasks while adhering to biological constraints. Researchers are exploring hybrid approaches that integrate bio-inspired principles with traditional deep learning techniques, such as combining spiking neural networks (SNNs) with conventional architectures. In addition to that, optimizing and developing specialized hardware such as neuromorphic chips is absolutely critical for enabling efficient event-based computation. These chips aim to emulate biological neural circuits but require further development to improve scalability, integration with existing systems, and support for complex dynamics necessary for advanced learning tasks.

Another key research direction involves developing biologically plausible methods for learning temporal sequences, as biological networks excel at processing time-dependent information. This includes leveraging the properties of SNNs and neuromorphic computing to naturally accommodate temporal dynamics. Researchers are also investigating how biological systems balance short-term and long-term memory, aiming to replicate these mechanisms in artificial systems for robust temporal learning. Furthermore, efforts are being made to optimize neural circuit dynamics by mimicking biological processes like synaptic plasticity and homeostasis, which enhance energy efficiency, learning speed, and adaptability. The development of fully biologically plausible algorithms is another critical area, focusing on replicating local learning rules, Hebbian plasticity, and the balance of excitatory and inhibitory signals while adhering to constraints like limited energy resources and real-time processing. These research directions collectively aim to bring biologically plausible learning and gradient descent methods closer to practical applications.

6 Conclusion

This review provides a critical analysis of backpropagation and highlights the significant advancements and ongoing challenges in the development of biologically plausible gradient descent algorithms such as Hebbian Learning, STDP, Feedback Alignment, Predictive Coding, and Target Propagation. We evaluated the biological plausibility of each method using three criteria and compared them against their most viable counterparts. We find that Hebbian Learning and STDP are the most biologically plausible algorithms; however, they fail to converge on the CIFAR-100 dataset. Methods such as predictive coding and the Forward-Forward algorithm most closely resemble brain activity patterns, at the cost of computational complexity. Energy-based learning and the Forward-Forward algorithm show great promise, though both struggle to fully replicate the spiking behavior of biological neurons and the unsigned nature of biological errors, which limits their effectiveness in mimicking real neural processes. By outlining the drawbacks of each method and suggesting future research directions, we aim to inspire

further innovations that deepen the connection between neuroscience and machine learning, ultimately leading to more effective and biologically inspired learning algorithms.

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